

Understanding the Primary Socioeconomic Drivers of State Test Scores, Grades 3-8

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Introduction

Standardized test scores are one of the statistics available for nearly every public school district in the country, making them a common measure to assess whether our education system is serving students well. Recent reporting on a generation-long decline in reading scores, paired with proposed cuts to federal Department of Education funding, has drawn increased attention to the matter of where funding should be allocated to improve test scores and more generally educational outcomes. Before discussing funding allocation, policymakers require a clear understanding of the data itself and the potential shared characteristics among school districts with lower test scores.

Socioeconomic status is a characteristic consistently mentioned in this conversation. Built off the correlations indicating that districts with higher incidents of poverty, and less educational attainment among adults tend to have lower-performing school test scores. But socioeconomic status bundles together several distinct conditions: income, education, employment, family structure, and use of public assistance (such as SNAP benefits utilization, etc.). Understanding and quantifying the relationship between socioeconomic status indicators and district state test scores with increased specificity is crucial when attempting to improve outcomes. A policymaker deciding how to allocate funding should understand not just that socioeconomic status is important, but which dimensions of it are most strongly associated with test performance, and how large those associations are in practice.

We approach this question as data consultants for a coalition of school boards seeking to better understand the primary drivers of achievement gaps in their districts. Our research question is purely descriptive: what is the relationship between district-level socioeconomic characteristics and average student test performance across U.S. school districts? We do not claim that any of these characteristics cause test scores to rise or fall, only that describing the strength and shape of these relationships is a necessary first step before anyone can design a targeted, evidence-based funding policy or a causal study of one.

Data

Source and Provenance

We use the Stanford Education Data Archive (SEDA), version 6.0 (Reardon et al., Stanford University), a publicly available dataset built from over 480 million standardized test records submitted by U.S. states under federal reporting requirements. SEDA converts each state's own assessments onto a common scale by linking them to the National Assessment of Educational Progress (NAEP), producing test scores that are comparable across states and years. We combine two SEDA files, both at the school-district level:

- `seda_geodist_annual_cs_6.0` (achievement): average math and reading/language-arts performance for grades 3-8,

- **seda_cov_geodist_annual_6.0** (covariates): district-level socioeconomic and demographic characteristics compiled from the American Community Survey and other federal sources.

For additional information regarding the methodology the SEDA utilized in collecting this data, as well as the files themselves, please visit their website at: <https://edopportunity.org/opportunity/data/downloads/#documentation-6>

Data Wrangling

From the achievement file, we keep only the **all students** aggregate (filtering on col **subcat = all**), which reports the achievement data in aggregate instead of by subgroups such as race or gender. Furthermore for both our achievement file and covariate file we will filter for **year = 2018**, as this is the most recent pre-pandemic year available to completely avoid any issues that may have resulted from COVID. Resulting in each row representing one achievement observation per district for 2018. We then join our data to the 2018 covariate file using the district identifier **sedalea**, which is common to both files. Districts that appear in only one file are dropped via an inner join, and rows missing any of our six predictor variables are also dropped.

Before dropping missing values, the 2018 achievement file has 11,139 district-level **all students** observations and the 2018 covariate file has 12,690. The inner join drops **1,551 covariate districts (12.2%)** that have no 2018 achievement estimate in any subgroup, which is SEDA’s suppression rule for districts with too few tested students. The loss runs one way only: every district with an achievement estimate has a matching covariate record, so the join itself discards nothing from the achievement side. A further **95 matched districts** are missing one or more of our six socioeconomic predictors and are dropped as well, leaving a population dataset of **11,044** observations, one per school district. (For reference, 324 districts in the covariate file carry no socioeconomic data at all; 95 of those are among the 11,139 that matched, and are the ones removed here.) As discussed in our previous proposal, districts are not drawn independently, districts neighboring each other likely have similar labor markets, demographics, and state funding resources which would naturally result in iid being violated. We will address this directly with a random sampling method rather than fitting on the full population with a resulting sample size of 1,000.

Operationalization

Outcome (Y): District test performance, **cs_mn_avg_ol** — a standardized effect size measuring how far a district’s average test score falls from the national mean, in national standard-deviation units. It is metric and pools math and reading/language-arts performance across grades 3–8.

Predictors (X), all district-level, metric:

Variable	Column	Definition
Income	lninc50all	Log of median household income
Educational attainment	baplusall	% of adults with a bachelor’s degree or higher
Poverty	povertyall	% of residents below the poverty line
Unemployment	unempall	% unemployment rate
SNAP receipt	snapall	% of households receiving SNAP benefits
Single motherhood	single_momall	% of households headed by a single mother

Data Prep

Sampling Methodology

Data & Sampling

Our data comes from the 2018 wave of the Stanford Education Data Archive (SEDA), a nationally comprehensive source that provides district-level academic performance and demographic information for public school districts across the United States. We work with the all-subgroups population file, which gives us one record per district, and restrict the frame to districts with positive enrollment and complete data on all variables we intend to analyze. That cleaning step leaves us with a well-defined population of districts before we draw a single observation.

The variables we retain reflect the core question. Our outcome is each district's average test score composite. Our socioeconomic predictors — bachelor's degree attainment, poverty rate, unemployment, SNAP receipt, single-mother household share, and log median income — are kept as separate measures rather than collapsed into a single index, precisely because we want to understand which dimensions of socioeconomic status carry the most weight. We also retain total enrollment and race and ethnicity shares (white, Black, Hispanic, Asian, and Native American) as contextual controls.

From that cleaned frame we draw a simple random sample of 1,000 districts without replacement. SRS is the right choice here because it satisfies the IID assumption directly: every district is selected independently and with equal probability, with no clustering or stratification introducing dependence into the sample. A fixed random seed ensures the draw is reproducible.

We then immediately split the sample before conducting any analysis. Three hundred districts (30%) are assigned to an exploration set; the remaining 700 (70%) are held as a confirmation set. All variable selection, distributional checks, and collinearity diagnostics are carried out on the exploration set only. The confirmation set does not inform any modeling decisions — it is reserved entirely for final estimation. This discipline matters because allowing the same data to guide both specification and estimation inflates the risk of finding patterns that do not replicate. Before writing any output, five automated checks verify that the sample is correctly sized, contains no missing values on analysis variables, includes only districts with positive enrollment, has unique district identifiers throughout, and that the exploration and confirmation counts sum correctly to 1,000.

Operationalization and data visualisation

Operationalization: evidence for the variable choices

This section supplies the evidence behind the variable choices summarised in the Operationalization subsection above; the two should be merged before submission rather than both appearing. Two concepts have to become variables: *student achievement* and *socioeconomic status*. Neither mapping is obvious, and both admit alternatives we rejected for stated reasons.

The outcome: district test performance

We operationalise achievement as `cs_mn_avg_ol` a district's mean score across math and reading, grades 3–8, expressed in standard deviations from the national average.

SEDA offers two candidate outcome columns. We use `cs_mn_avg_ol` (the ordinary least-squares estimate) rather than `cs_mn_avg_eb` (an empirical-Bayes estimate that shrinks small districts toward the state mean). Shrinkage would improve precision but at the cost of *deliberately* pulling small districts toward a common

value exactly the between-district variation we are trying to describe. For a descriptive study the unshrunk estimate is the honest choice.

Statistic	Value	Reference
Districts	300	–
Distinct values	300	300 if no ties
Districts within 0.05 of the minimum	1	high value = floor effect
Districts within 0.05 of the maximum	3	high value = ceiling effect
Mean	-0.005	–
SD	0.380	–
Min	-1.112	–
25th pct	-0.265	–
Median	-0.027	–
75th pct	0.230	–
Max	1.134	–
Skewness	0.193	0 if symmetric
Kurtosis	3.210	3 if normal

The table shows the variable needs no transformation. All 300 districts have a distinct score and none pile up at either extreme, so nothing is censored. Skewness 0.19 and kurtosis 3.21 sit near their symmetric and normal reference values. Symmetry is not an OLS requirement the large-sample model assumes nothing about the marginal distribution of Y but it does mean nothing suggests the outcome behaves multiplicatively, so **we leave Y untransformed**. The scale is already interpretable: a coefficient reads directly as a change in standard deviations.

One limitation to state plainly. `cs_mn_avg_o1` is an *estimate*, and SEDA publishes a standard error for each district. About 1.9% of the outcome’s variance is measurement error rather than genuine difference between districts, and that error is concentrated in the smallest districts (the correlation between a district’s standard error and its log enrolment is -0.79).

The predictors: socioeconomic status as six measures, not one index

Socioeconomic status could be operationalised as a single composite SEDA ships one, `sesall`. We deliberately do not use it. The research question asks *which dimensions* of socioeconomic status carry the most weight, and a composite answers that question by erasing it. So we keep six separate measures: log median income, adult bachelor’s-degree share, poverty rate, unemployment, SNAP receipt, and single-mother household share.

That choice has a cost, and it is worth quantifying rather than glossing.

Table 1: Variance inflation factors, all six predictors entered together.

Variable	VIF
Households below the poverty line (%)	6.15
Log median household income	6.11
Households receiving SNAP (%)	4.81
Adults with a bachelor’s degree or higher (%)	2.98
Households headed by a single mother (%)	2.72
Unemployment rate (%)	1.89

Predictor	Slope alone	p-value alone	Slope with all 6	p-value with all 6
Households receiving SNAP (%)	-3.501	0	0.199	0.636
Log median household income	0.821	0	0.025	0.803

The six measures overlap heavily poverty correlates with SNAP receipt at 0.85 and with log income at -0.82. The variance inflation factors show the consequence, and the second table makes it concrete: SNAP receipt and log income are both strong, highly significant predictors on their own, but entered alongside the other four SNAP **flips sign** and loses all significance while income collapses toward zero.

Neither variable has stopped mattering; each has been absorbed by its correlates. We therefore **exclude SNAP receipt and log median income** from the preferred specification not because they are unimportant, but because their coefficients in a six-predictor model would not mean what a reader would take them to mean. Poverty is retained over SNAP because it is the more widely understood measure; education and poverty together carry most of what income would have contributed.

Observations removed

Table 2: Observations removed at each stage of the pipeline, with reasons. Generated by the cleaning code, not transcribed.

Step	Rows in	Rows out	Rows dropped	%	Reason
Achievement rows to district rows	110395	11139	99256	89.9	Kept subcat/subgroup ‘all’ and gap == 0. The other rows are subgroup means and subgroup gaps, not district means. A reshape, not a data loss.
Covariate districts to matched districts	12690	11139	1551	12.2	Districts with covariates but no 2018 achievement estimate. SEDA suppresses districts with too few tested students.
Matched districts to complete records	11139	11044	95	0.9	Missing one or more socioeconomic covariates (listwise deletion).
Frame to analysis sample	11044	1000	10044	90.9	Simple random sample without replacement, set.seed(42).
Analysis sample to exploration set	1000	300	700	70.0	Confirmation set held out; not opened until final estimation.

The first row is a reshape, not a loss: the achievement file stacks subgroup means and achievement gaps alongside district means, and only the district rows are our unit of observation. The drop that matters is the **1,551 districts SEDA suppressed** for having too few tested students, plus a further 95 missing one or more covariates.

Because SEDA still publishes covariates for the suppressed districts, we can check whether that loss is systematic rather than assuming it.

Measure	Retained	Suppressed
Districts	11139	1551
Median enrolment	561	391
25th percentile of enrolment	219	56
Mean poverty rate	12.7%	12.1%

Measure	Retained	Suppressed
Mean bachelor's+ share	24.3%	28.3%
Share rural	52.8%	59.6%

This corrects an assumption it would have been easy to make. The suppressed districts are genuinely smaller markedly so at the lower quartile and somewhat more rural. But they are **not** poorer or less educated; on both measures they look slightly better off than the districts retained. The scope condition is therefore narrower than a vague caveat would suggest: **our results describe districts large enough for SEDA to publish a score**, and what is missing is very small, somewhat rural districts rather than a systematically disadvantaged tail.

The relationship between district poverty and test performance

This is the report's central exhibit: the outcome plotted against the single predictor that carries most of the story, with the fitted model's predictions overlaid.

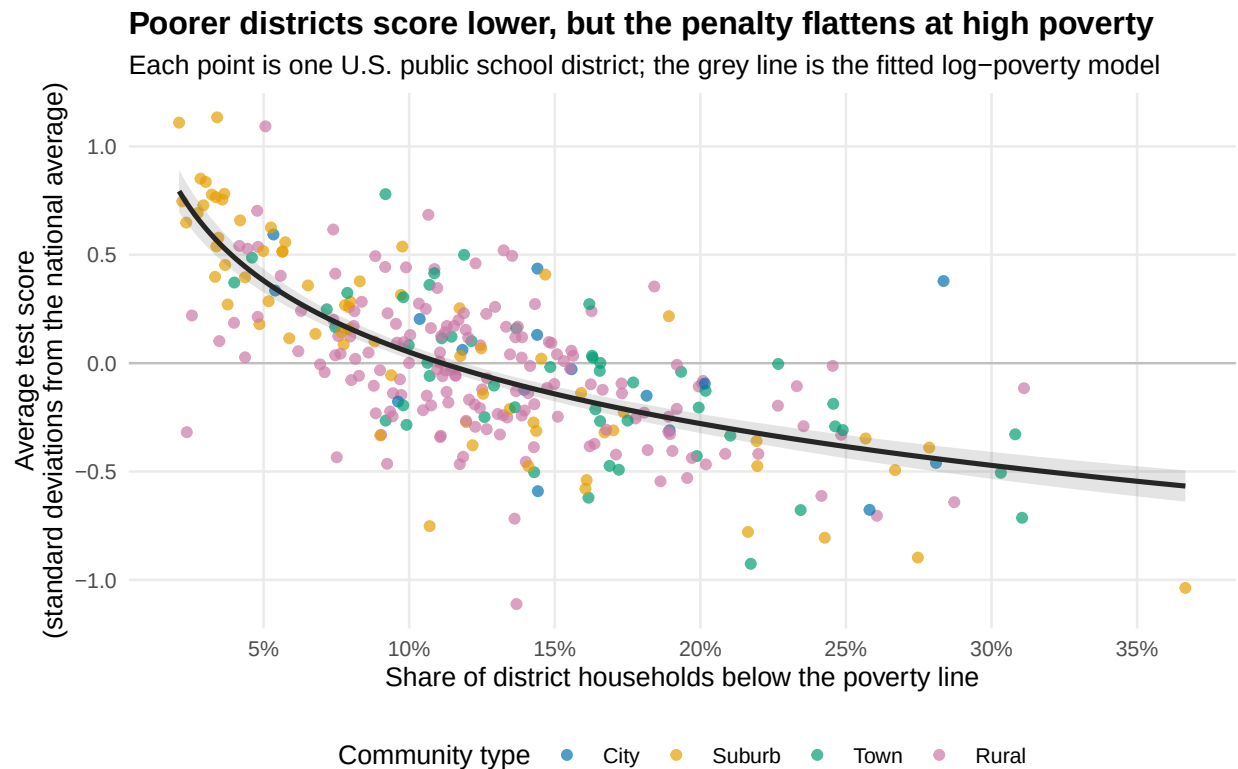


Figure 1: District poverty and average test performance, 300 randomly sampled U.S. school districts, 2018. Each point is one district. The solid line is the fitted relationship from a model of test score on log poverty rate; the shaded band is its 95% confidence interval. Districts are coloured by community type. The relationship is steep among low-poverty districts and flattens as poverty rises.

the figure below displays all 300 sampled districts. The relationship is strongly negative but plainly **not a straight line**: the fitted curve falls steeply across the low-poverty range and flattens beyond roughly 20% poverty. Suburban districts cluster at the top left low poverty, high scores while rural districts, the most

numerous group at 163 of 300, spread across the full range. Community type is shown for context only; it is not a term in any model.

Poverty rate	Predicted score	Change from row above
5%	0.38	NA
10%	0.05	-0.33
15%	-0.14	-0.19
20%	-0.28	-0.14
30%	-0.47	-0.19

The table translates the curve into districts a reader can picture. Moving a hypothetical district from 5% to 15% poverty is associated with a fall of about 0.52 standard deviations; the *larger* move from 15% to 30% is associated with only about 0.33. Ten points of poverty are worth roughly twice as much at the bottom of the range as fifteen points are at the top.

That asymmetry is the descriptive finding, and it is why the poverty rate enters the model in logs. A straight line would force both moves to cost the same and would misrepresent what the data shows.

Appendix: specifications tried

Table 3: Three functional forms, tested on both variables where curvature was plausible. Higher R-squared and lower residual SE are better.

Variable	Form	R-squared	Residual SE
Poverty rate	Linear	0.4481	0.2831
Poverty rate	Quadratic	0.4834	0.2744
Poverty rate	Log	0.4924	0.2715
Bachelor's+ share	Linear	0.4592	0.2803
Bachelor's+ share	Quadratic	0.4595	0.2807
Bachelor's+ share	Log	0.4372	0.2859

Specification	What we learned
Y on poverty rate, untransformed	Fits, but the residual pattern and the loess curve both show the straight line is wrong at the extremes.
Y on poverty rate plus a squared term	Curvature is real and strongly significant, so the linear form is rejected. But a fitted parabola eventually turns and predicts higher scores at very high poverty.
Y on log(poverty rate)	Best fit of the three, monotone across the whole range, and interpretable as diminishing returns. Adopted.
Y on bachelor's share, log and squared variants	Neither improves on the linear form, so education enters untransformed. Curvature is not a general feature of these predictors.
Y on all six socioeconomic measures	SNAP receipt flips sign and log income collapses. The measures are too collinear to interpret individually.
Y on log(poverty), education, unemployment, single motherhood	Retains the explanatory power of the six-predictor model with variance inflation near or below three. Preferred specification.

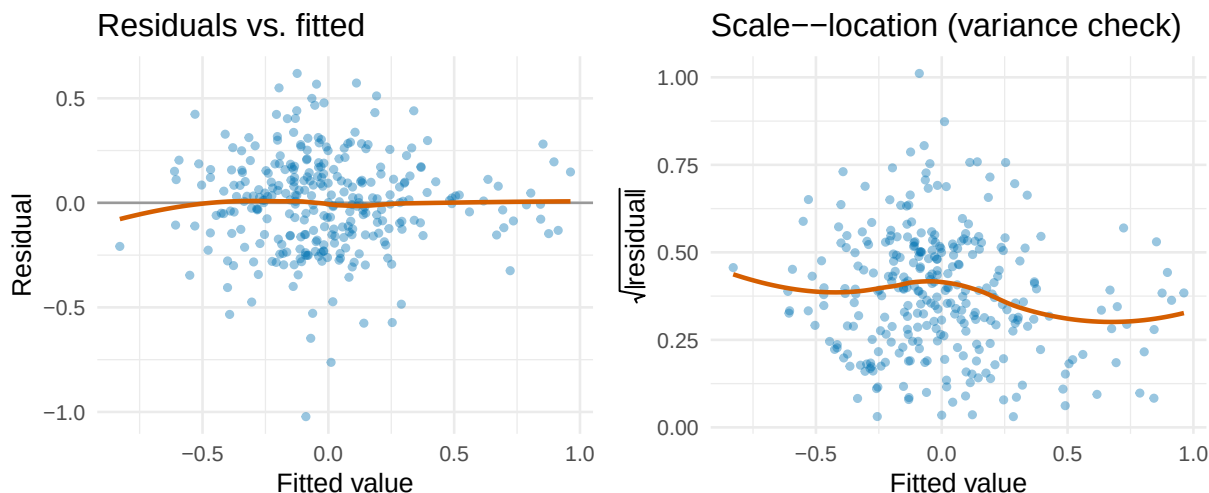


Figure 2: Residuals against fitted values for the proposed four-variable model. The local fit (orange) is close to flat with no funnelling, consistent with a correctly specified conditional mean and roughly constant error variance.

The comparison table above is the evidence for rows one to four: for poverty the log form fits best, while for the bachelor's-degree share the linear form already does, so only poverty is transformed.

the residual plot below is the required residuals-versus-fitted plot for the preferred specification. The local fit is close to flat with no systematic curvature, which is consistent with a correctly specified conditional mean had the poverty rate been left on its raw scale, the curvature documented above would appear here as a parabolic band.

Model Building

All model choices below are based only on the exploration data. We do not use the confirmation set until the very end, so none of the decisions will introduce bias in the final results.

`## Exploration set: 300 districts`

We also create three unique models

Model 1 showcases a comparison between test scores vs. poverty rate only. Consider this our starting point because poverty is the measure most used in policy discussions

Model 2 will add the other five factors (income, education, employment, SNAP, family structure); This shows us how much of poverty's impacts on its own then with the other factors

Model 3 adds a squared poverty term so that we can check if the relationship between the two is a straight line or curves at high levels of poverty

We use robust standard errors instead of regular ones since this type of district-level data commonly has uneven variance.

This chunk builds the side-by-side regression table comparing for model 1,2,and 3.

Table 4: Regression Results (Exploration Set)

	<i>Dependent variable:</i>		
	Test Performance (SD units from national mean)		
	(1)	(2)	(3)
Poverty Rate	−4.02*** (0.29)	−1.24* (0.57)	−2.59* (1.18)
Poverty Rate Squared			4.14 (3.09)
Log Median Income		0.03 (0.12)	−0.01 (0.12)
% Bachelor's Degree+		1.30*** (0.18)	1.26*** (0.19)
Unemployment Rate		−2.30* (1.10)	−2.38* (1.11)
% SNAP Recipients		0.20 (0.48)	0.14 (0.48)
% Single-Mother Households		−1.19*** (0.36)	−1.23*** (0.35)
Constant	0.51*** (0.04)	−0.16 (1.27)	0.32 (1.32)
Observations	300	300	300
R ²	0.45	0.63	0.63
Adjusted R ²	0.45	0.62	0.62

Note:

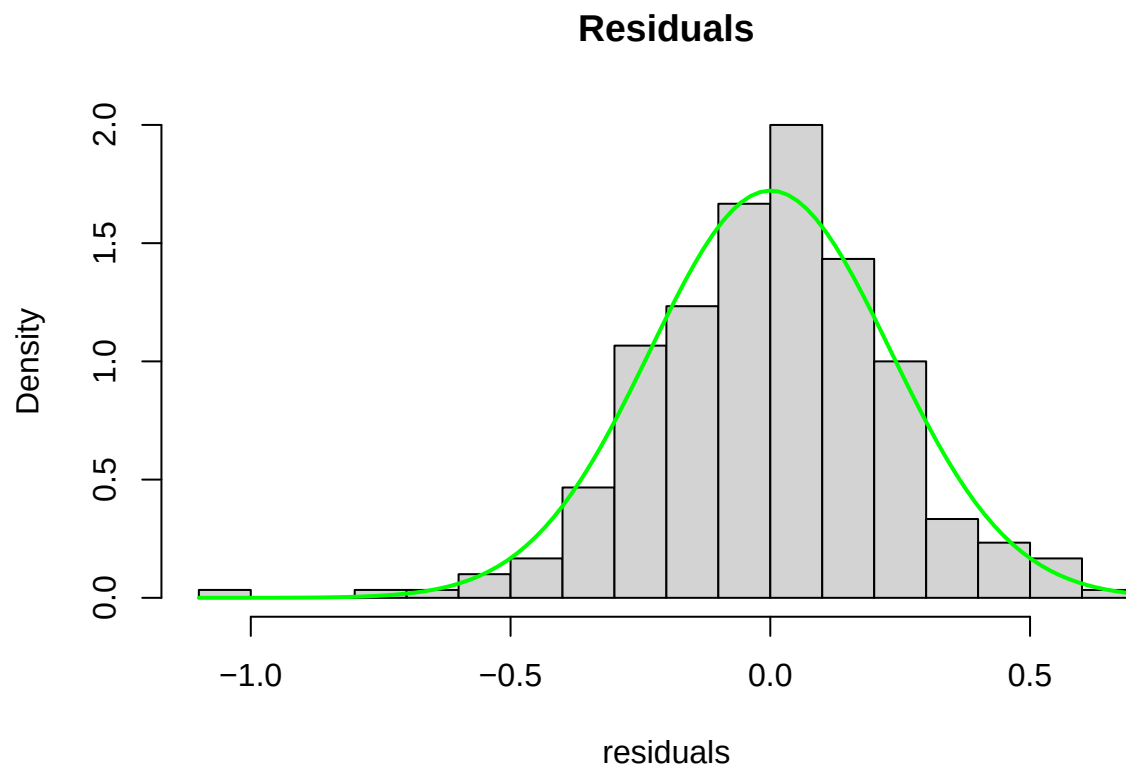
*p<0.05; **p<0.01; ***p<0.001

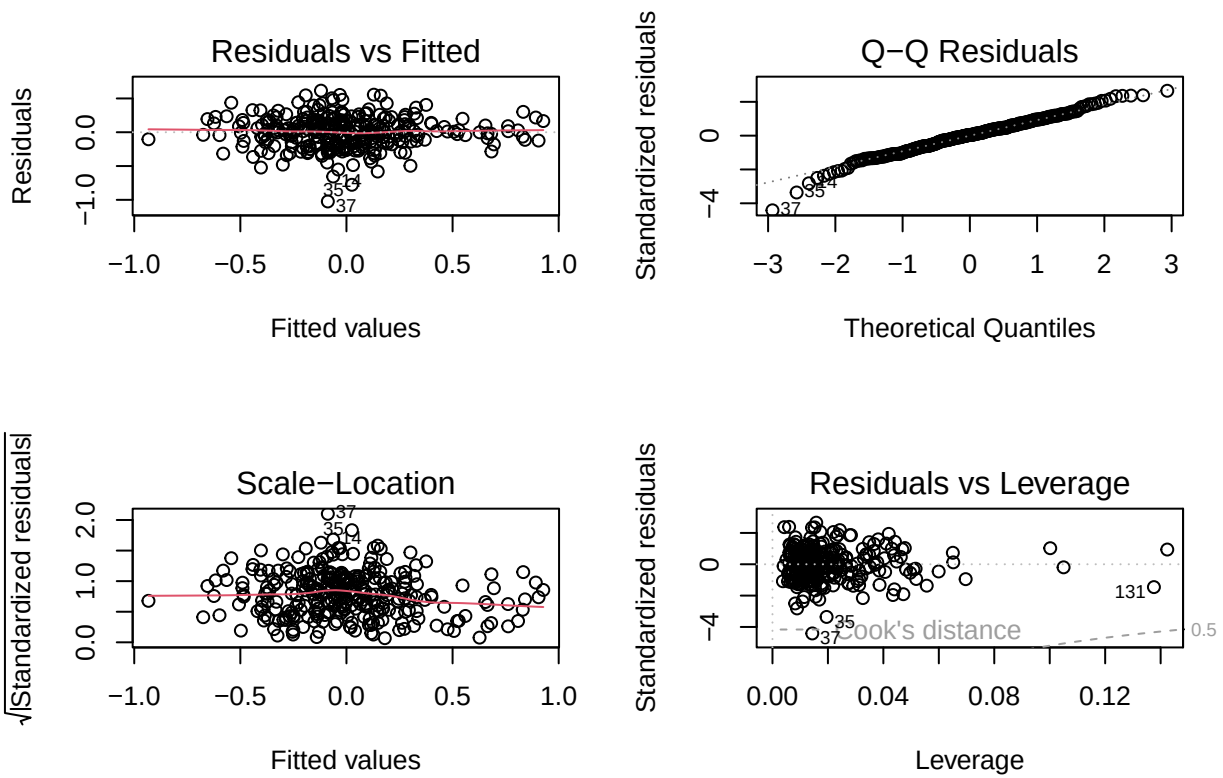
Model Assumption checks

```
##  
## studentized Breusch-Pagan test  
##  
## data: m4  
## BP = 6.228, df = 5, p-value = 0.2847
```

```
## [1] 1.234734
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals  
## W = 0.98564, p-value = 0.004327
```





```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.12199   0.05749   2.1220  0.034676 *
## povertyall  -1.32006   0.42094  -3.1360  0.001886 **
## baplusall    1.33277   0.11389  11.7021 < 2.2e-16 ***
## unempall    -2.28260   1.09005  -2.0940  0.037113 *
## snapall      0.20065   0.47915   0.4188  0.675703
## single_momall -1.19341  0.35825  -3.3312  0.000975 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```